

HYBRID DOCUMENTATION APPROACH

HealthConnect Experience Lab — Data Analytics Track | Week 4

Documentation Architecture for the Week 4 Analytical Foundation

DOCUMENTATION DECISION

Initial Analysis Document = Main Track-Specific Week 4 Output | Technical Data Inspection Notebook = Supporting Technical Evidence

1. PURPOSE & RATIONALE

AnalystLab requires an Initial Analysis Document or Notebook for the Data Analytics track. For Week 4, the selected approach combines both formats while assigning each a distinct role: the Initial Analysis Document serves as the principal analytical communication output, while the Technical Data Inspection Notebook provides the supporting evidence, reproducibility, and traceability behind the technical observations.

This architecture separates analytical communication from technical validation and preserves the intended Week 4 boundary: understanding the problem and resources, assessing the data, defining the analytical direction, and planning the next stages rather than prematurely executing the full analysis or building a final dashboard.

2. INITIAL ANALYSIS DOCUMENT | MAIN ANALYTICAL OUTPUT

Role: Present a clear, structured, and professional view of the HealthConnect business problem, the available appointment data, the questions that should be investigated, and the proposed analytical approach.

- Project context and HealthConnect business problem.
- Week 4 objective and analytical scope.
- Data Dictionary Review and Dataset Overview.
- Initial Data Quality Assessment.
- Relevant Variables Identification and analytical architecture.
- Relevant Business Questions.
- Analytical Hypotheses — Value-Added.
- Three Potential KPIs linked to the Business Questions.
- Initial Analysis Approach.
- Assumptions, Limitations, Risks & Dependencies.
- Week 4 conclusion and preparation for the next project stage.

Communication principle: The document should explain what is known at this stage, what still needs to be investigated, why those investigations matter, and how they are expected to be conducted — without presenting untested relationships as findings.

3. TECHNICAL DATA INSPECTION NOTEBOOK | SUPPORTING EVIDENCE

Role: Make the technical statements in the main document verifiable and reproducible from the raw dataset.

- Dataset loading, dimensions, columns, and data types.
- Dataset ↔ Data Dictionary alignment checks.
- Missing-value and duplicate checks.
- Identifier uniqueness and categorical/numeric validity checks.
- Cross-field consistency checks.
- Consistency checks across repeated patient_id values.
- Consolidated Data Quality Issue Register.
- Documented technical results and their analytical implications.

4. TRACEABILITY STANDARD

Every material factual statement in the Initial Data Quality Assessment should be traceable to a reproducible notebook check. The notebook will therefore follow a consistent evidence sequence:

PURPOSE → CODE → RESULT → INTERPRETATION

5. WEEK 4 SCOPE BOUNDARY

The notebook remains an initial inspection and technical validation tool. It must not become an EDA that already answers the Business Questions or produces final recommendations.

IN SCOPE — WEEK 4	OUT OF SCOPE — WEEK 4
Structure and data-quality inspection	Answering the Business Questions
Missing values, duplicates, and type checks	Testing the analytical hypotheses
Validity and consistency controls	Producing final analytical visualizations
Documentation of limitations	Building a complete dashboard
Planning the future analysis	Issuing final recommendations

6. HOW THE TWO OUTPUTS WORK TOGETHER

RAW DATASET → TECHNICAL NOTEBOOK → VALIDATED EVIDENCE → INITIAL ANALYSIS DOCUMENT → FUTURE ANALYSIS

Notebook = Evidence Layer: demonstrates how technical observations were obtained and preserves reproducibility.
Initial Analysis Document = Analytical Communication Layer: converts validated observations into a structured understanding of the problem, Business Questions, potential KPIs, planned analysis, and interpretation boundaries.

7. WEEK 4 DELIVERABLE POSITIONING

OUTPUT	ROLE	POSITIONING
Initial Analysis Document	Main track-specific analytical communication output	REQUIRED
Technical Data Inspection Notebook	Technical evidence and reproducibility layer	SUPPORTING FILE
Week 4 Project Summary	Concise complementary summary	REQUIRED

8. FINAL METHODOLOGICAL DECISION

The HealthConnect Week 4 Data Analytics submission will use a hybrid documentation architecture in which the Initial Analysis Document is the principal track-specific output and the Technical Data Inspection Notebook is the supporting technical evidence layer.
This approach strengthens compliance, readability, technical traceability, and reproducibility while maintaining a clear separation between the Week 4 foundation stage and the analytical execution that will follow in subsequent stages.

INITIAL ANALYSIS DOCUMENT = MAIN ANALYTICAL OUTPUT
+
TECHNICAL DATA INSPECTION NOTEBOOK = SUPPORTING TECHNICAL EVIDENCE

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